University of BRISTOL

AVDASI-3: Sustainable Aviation & Ethics

Lecture-7: Sustainable Propulsion & Fuel Systems

Samudra Dasgupta

Senior Lecturer – CADE, University of Bristol

bristol.ac.uk

University of BRISTOL

Lecture-7: Sustainable Propulsion & Fuel Systems

7A: Gas Turbine Engines

Samudra Dasgupta

Senior Lecturer – CADE, University of Bristol

bristol.ac.uk

Performance & propulsion evolution

Aircraft	Payload	Range	Speed	Engine(s)	Engine Power / wt.
Wright Flyer – I (Flight # 4) 1900s	1 pax (Wilbur)	852 feet	48 km/h	1x IC engine	9 kW / 90 kg = 0.1 kW/kg
Antanov 225 Mriya 1980s	250 tons	4,500 km (loaded)	850 km/h (0.64M)	6x Progress D-18T TF engines	11,415 kW / 4100 kg = 2.8 kW/kg
B777-200LR 1990s	65 tons	15,843 km	950 km/h (0.84M)	2x GE90-115B TF engine	28,500 kW / 8300 kg = 3.4 kW/kg
Concorde 1960s	13 tons	7,222 km	2,180 km/h (2.04M)	4x RR/Snecma Olympus 693 TJ engines with AB	6300 kW / 1200 kg = 5.25 kW/kg

Elements of propulsion systems

Energy Source	Energy to Work Converter (Powerplant)	Thrust Producer

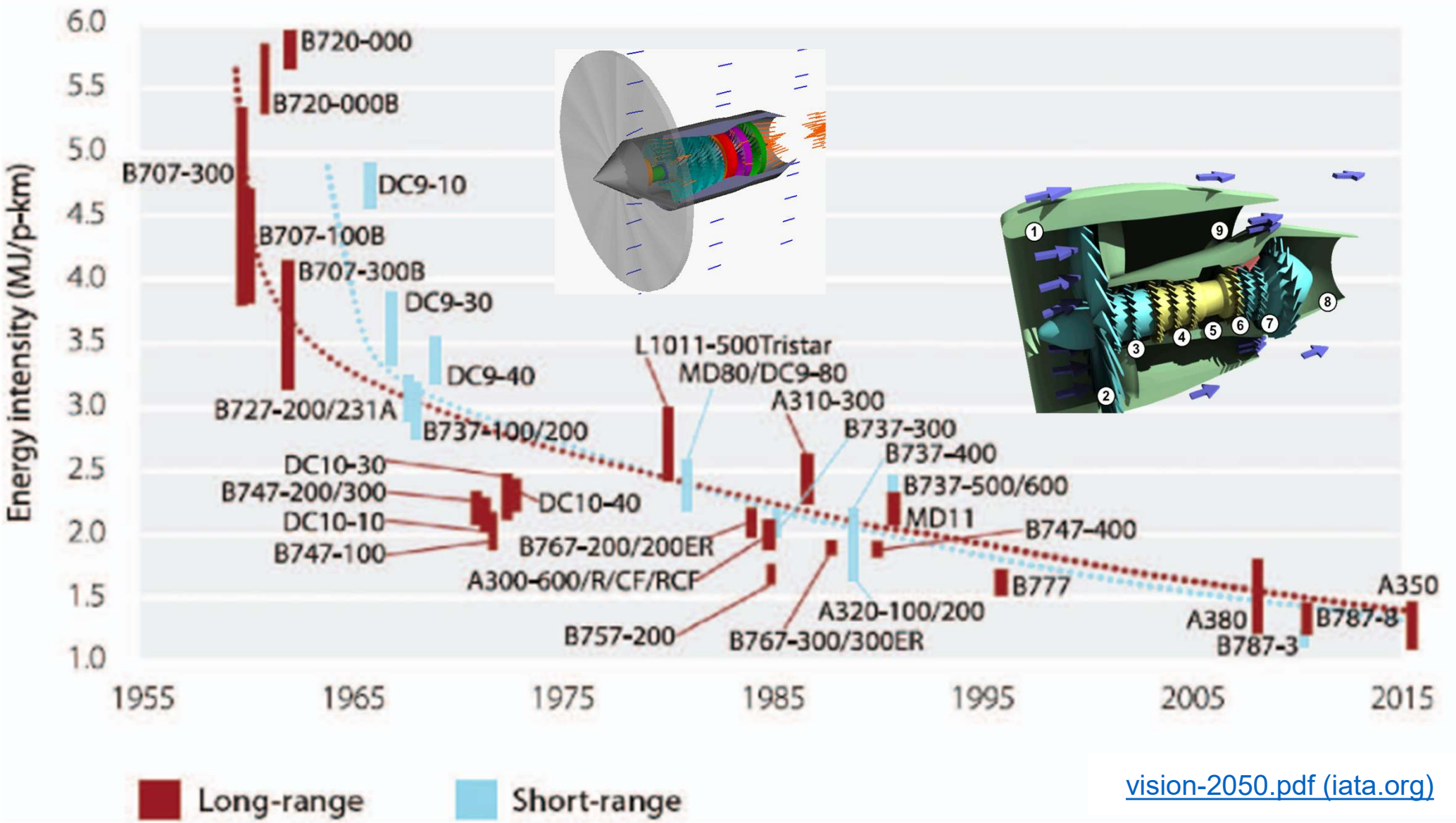
• Batteries
  • Ni-CAD: 0.2 MJ/kg
  • Li-ion: 1.2 MJ/kg
  • Li-S: 2.5 MJ/kg
• Fuel
  • Kerosene / SAF: 43 MJ/kg (high VED)
  • H_2: 120 MJ/kg (low VED)
  • Nuclear fuel?
  • Solar cells?
	• Fuel cells (350 W/kg)
	• IC / Piston engine (1-3 kW/kg)
	• Wankel / Rotary engine (3 kW/kg)
	• Electric motors (9 kW/kg)
	• Gas turbine core (25 kW/kg)
		• Propeller
		• Nozzle

Civil Aviation: Kerosene / SAF + Gas Turbine + Ducted Fan (Propeller + Nozzle)
General Aviation: Kerosene + IC Engine + Propeller
Combat A/c: Kerosene + Gas Turbine with AB + Nozzle
Small UAV: Battery + Electric Motor + Propeller
Electric A/c (UAM): HED Battery + Electric Motor + Propellers

University of BRISTOL

Turbofan efficiency

bristol.ac.uk

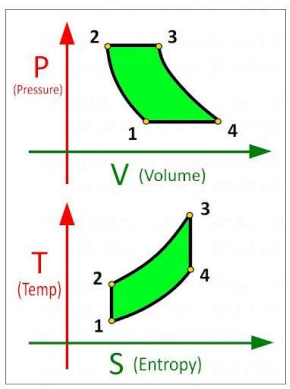


vision-2050.pdf (iata.org)

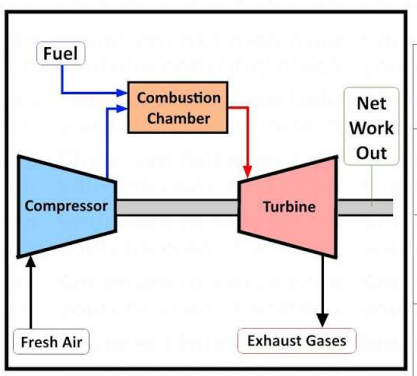
Thermodynamics of jet engine

- Brayton / Joule cycle
  - Compressor (driven by turbine) to compress fresh air
  - Fuel is burnt (in combustion chamber) to increase Temp.
  - Air expands, powers the turbines & accelerates backwards (Newton's 2nd Law)
  - Reaction thrust (Newton's 3rd Law)

PV & TS Diagram

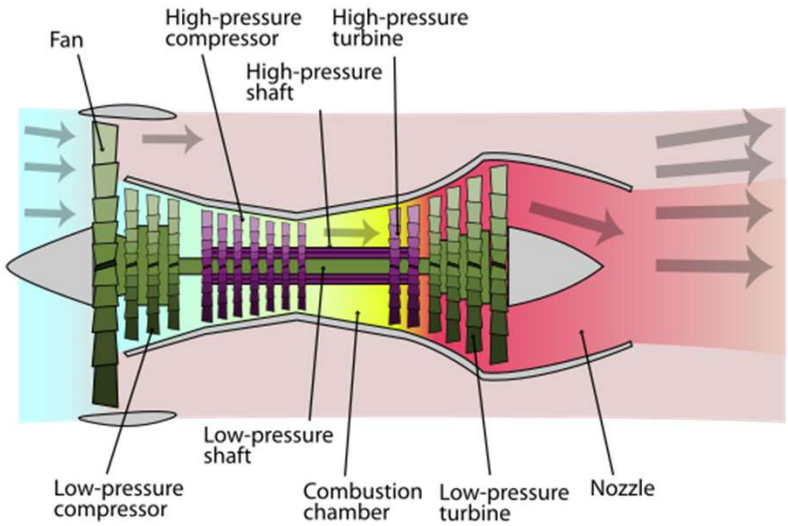


Gas Turbine

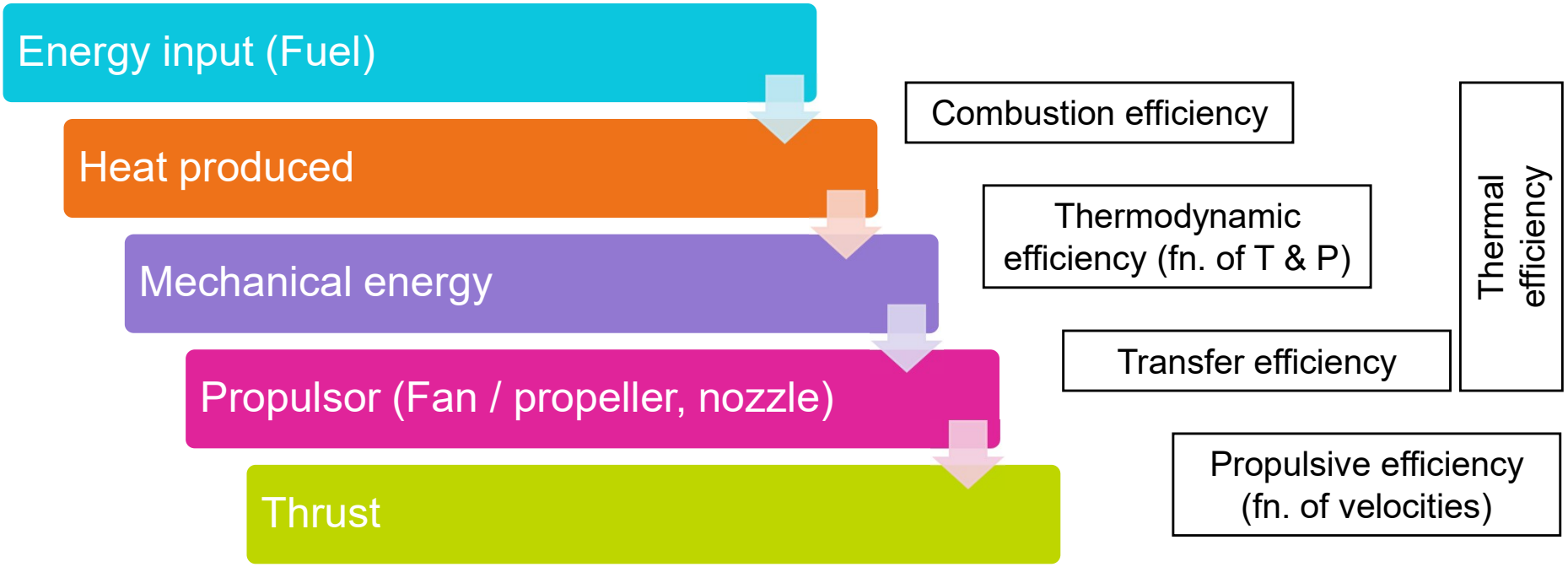


Thermodynamic Processes

Process 1 to 2 Isentropic Compression
Process 2 to 3 Isobaric Heat Addition
Process 3 to 4 Isentropic Expansion
Process 4 to 1 Isobaric Heat Rejection



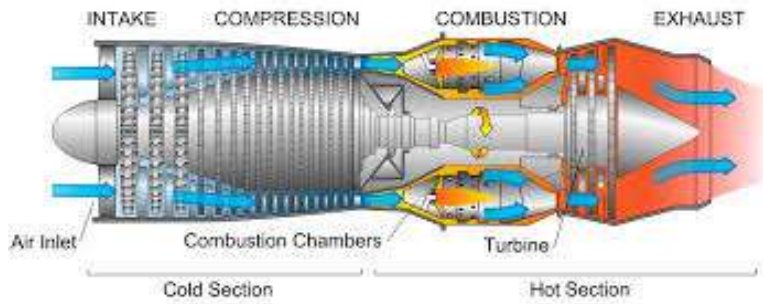
Gas turbine efficiencies



Physics of jet engine

$$\dot{m} = \frac{dm}{dt}$$

$$\xrightarrow[V_{in}]{\dot{m}}$$



$$\xrightarrow[V_{out}]{\dot{m}}$$

Thrust (T) = $\dot{m}$ ($V_{out} - V_{in}$)

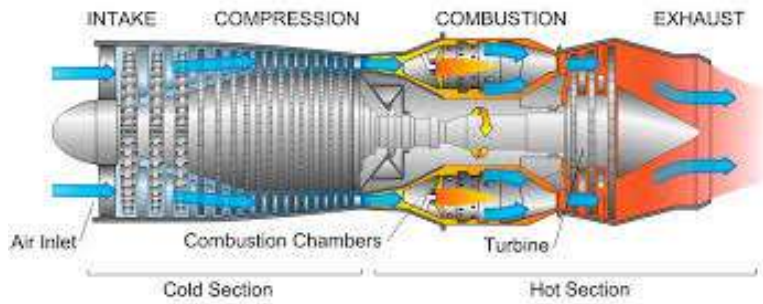
Power (P) = Work / t = (T x Distance) / t = T x V_{in} (speed of a/c = speed of air) = $\dot{m}$ ($V_{out} - V_{in}$) x V_{in}

ΔKE / t = ½ $\dot{m}$ ($V^2_{out} - V^2_{in}$)

Jet engine propulsive efficiency

$$\dot{m} = \frac{dm}{dt}$$

$\dot{m}$ → V_{in}



$\dot{m}$ → V_{out}

$$\text{Efficiency } (\eta) = \frac{\text{What I gain}}{\text{What I pay for that}} = \frac{\text{Power}}{\Delta KE/t}$$

$$\eta = \frac{\dot{m}\,(V_{out} - V_{in}) \times V_{in}}{\frac{1}{2}\,\dot{m}\,(V_{out}^2 - V_{in}^2)} = \frac{2V_{in}}{V_{out} + V_{in}}$$

$$\eta = \frac{2}{1 + (V_{out} / V_{in})}$$

$\eta \uparrow \Rightarrow V_{out} \downarrow \Rightarrow T = \dot{m}\,(V_{out} - V_{in}) \downarrow \Rightarrow$ **undesirable**

$\eta \uparrow$ **&** $T \uparrow \Rightarrow \dot{m} \uparrow$ **&** $V_{out} \downarrow \Rightarrow$ **desirable**

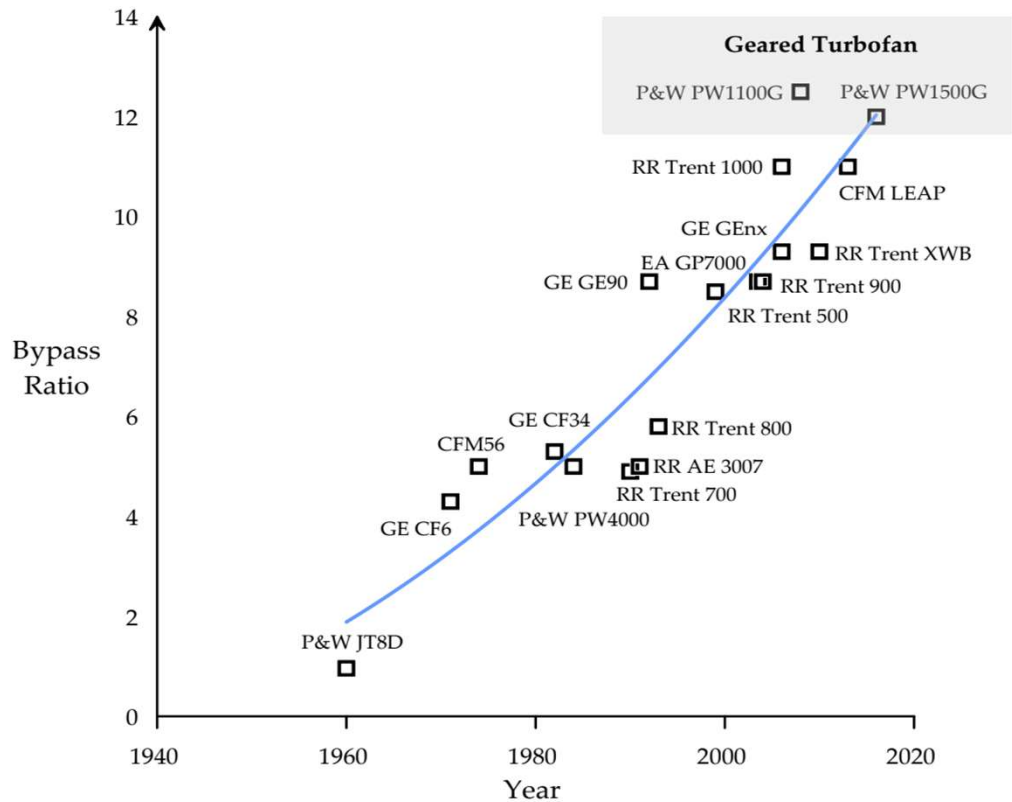
<u>Better to accelerate lot of air a little bit rather than a bit of air by a lot</u>

Engines are getting larger (high BPR)

Trends in efficiency parameters - BPR

By-pass Ratio (BPR)

- Improved aerodynamics
- Lighter & impact resistant materials
- Improved manufacturing

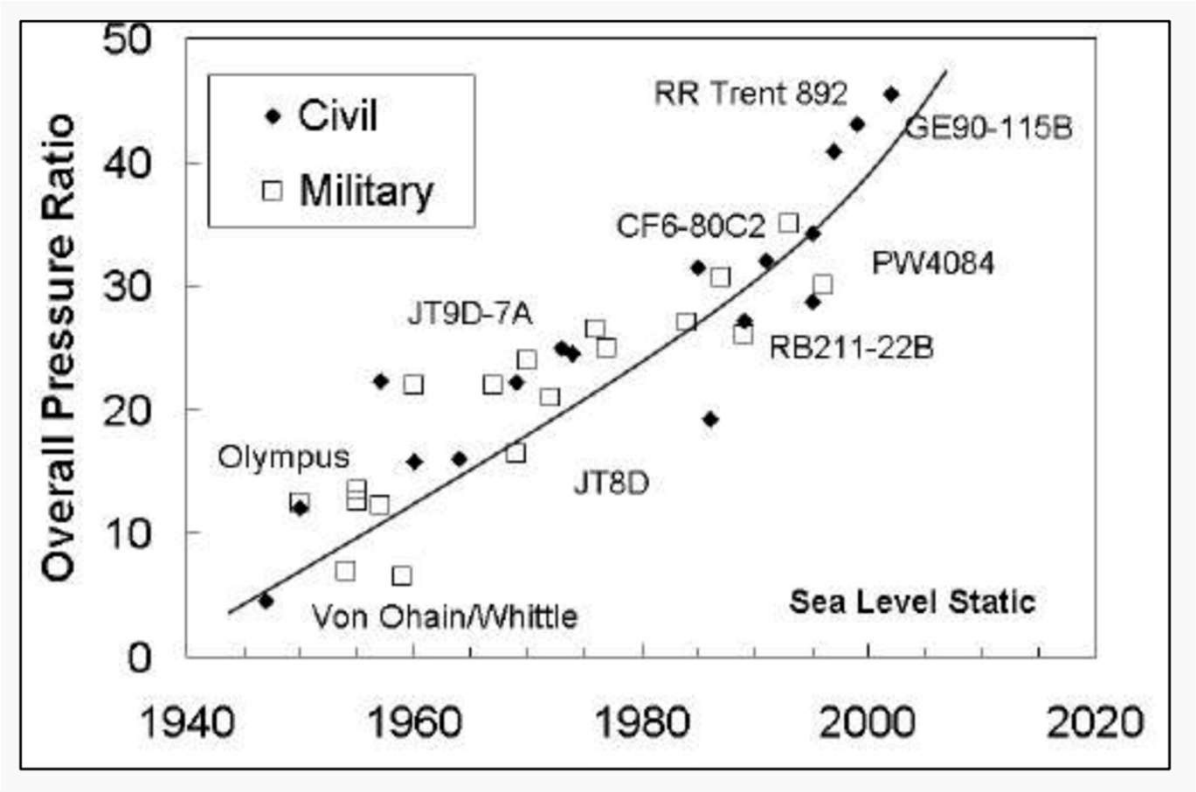


Bypass ratio - Wikipedia

Trends in efficiency parameters - OPR

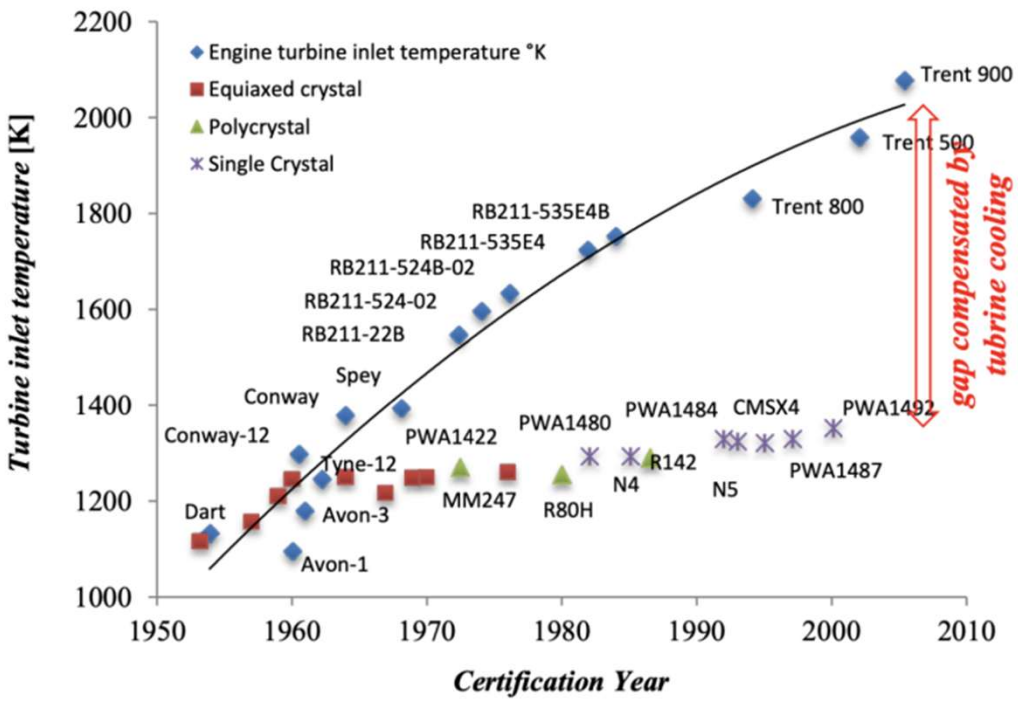
Overall Pressure Ratio (OPR)

- Improved aerodynamics
- Better / lighter materials
- Improved manufacturing tolerances



Jet Engines: Evaluation of Overall Pressure Ratio Advancements - O'Day Aviation Consulting

Trends in efficiency parameters - TIT

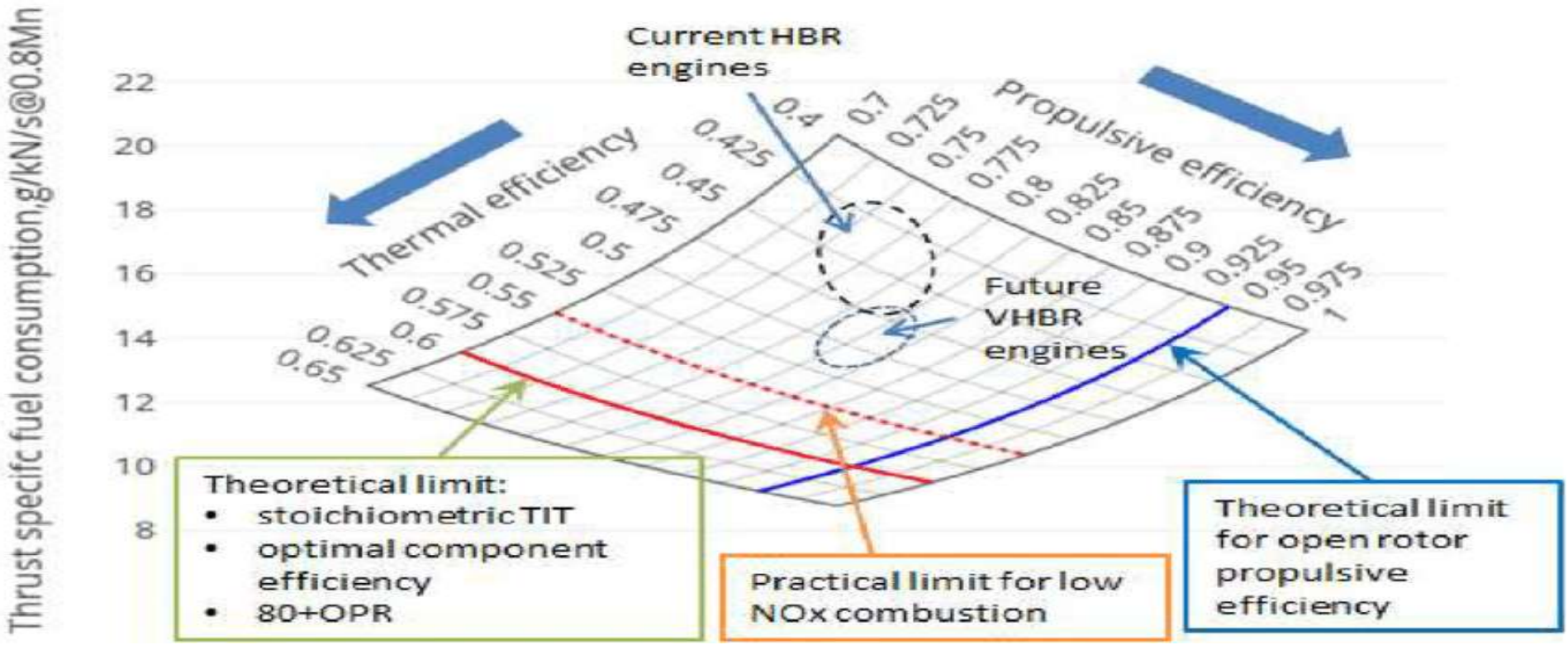


Jet Engines: Evaluation of Overall Pressure Ratio Advancements - O'Day Aviation Consulting

Turbine Inlet Temperature (TIT)

- Improved manufacturing (single crystals)
- Improved cooling effectiveness
- Creep resistant materials
- Thermal coatings

The efficiency limits of turbofan engines



(1) (PDF) A review of gas turbine engine with inter-stage turbine burner

University of BRISTOL

Lecture-7: Sustainable Propulsion & Fuel Systems

7B: Other Decarbonisation Options

Samudra Dasgupta

Senior Lecturer – CADE, University of Bristol

bristol.ac.uk

Elements of propulsion systems

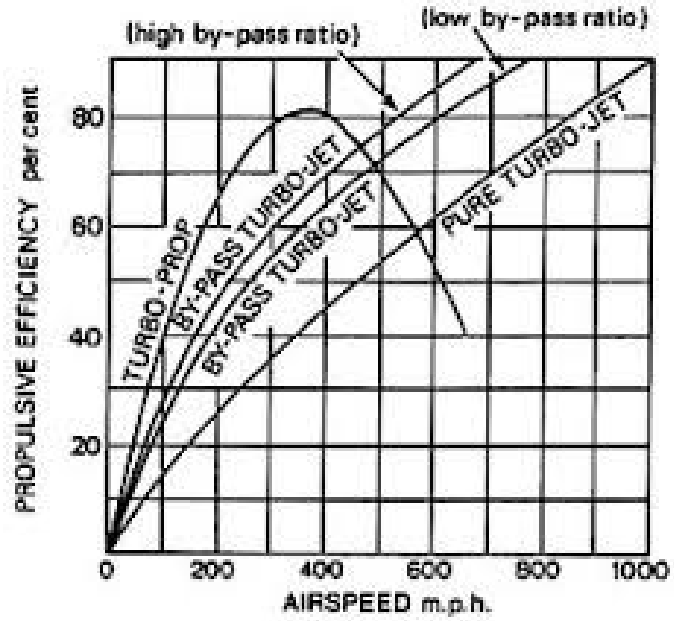
Energy Source	Energy to Work Converter (Powerplant)	Thrust Producer

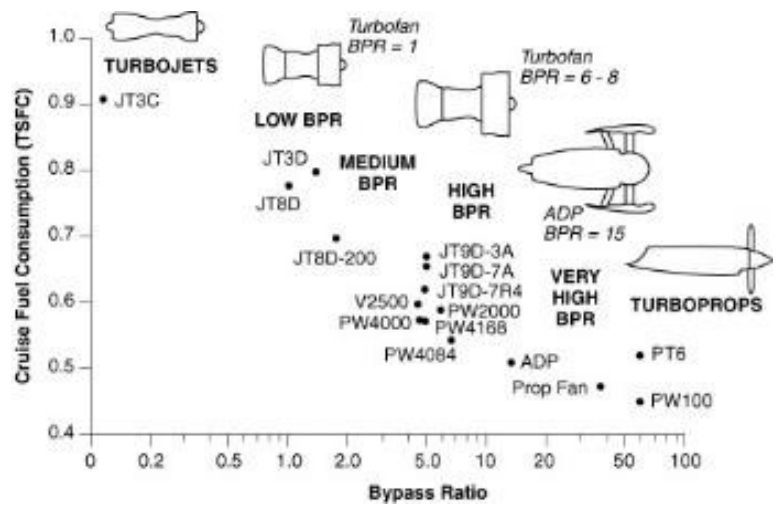
• Batteries
  • Ni-CAD: 0.2 MJ/kg
  • Li-ion: 1.2 MJ/kg
  • Li-S: 2.5 MJ/kg
• Fuel
  • Kerosene / SAF: 43 MJ/kg (high VED)
  • H_2: 120 MJ/kg (low VED)
  • Nuclear fuel?
  • Solar cells?
	• Fuel cells (350 W/kg)
	• IC / Piston engine (1-3 kW/kg)
	• Wankel / Rotary engine (3 kW/kg)
	• Electric motors (9 kW/kg)
	• Gas turbine core (25 kW/kg)
		• Propeller
		• Nozzle

Civil Aviation: Kerosene / SAF + Gas Turbine + Ducted Fan (Propeller + Nozzle)
General Aviation: Kerosene + IC Engine + Propeller
Combat A/c: Kerosene + Gas Turbine with AB + Nozzle
Small UAV: Battery + Electric Motor + Propeller
Electric A/c (UAM): HED Battery + Electric Motor + Propellers

Turboprops

- Not shrouded, very high BPR
- High tip losses at transonic regime
- Excellent at low Mach nos.
- Advantages
  - Most efficient ~ 90%
  - Compatible with electric propulsion
  - **Distributed propulsion systems**
- Challenges
  - Unsteady aerodynamic interactions
  - Noise & vibrations
  - System complexity & higher weight
  - Stability issues

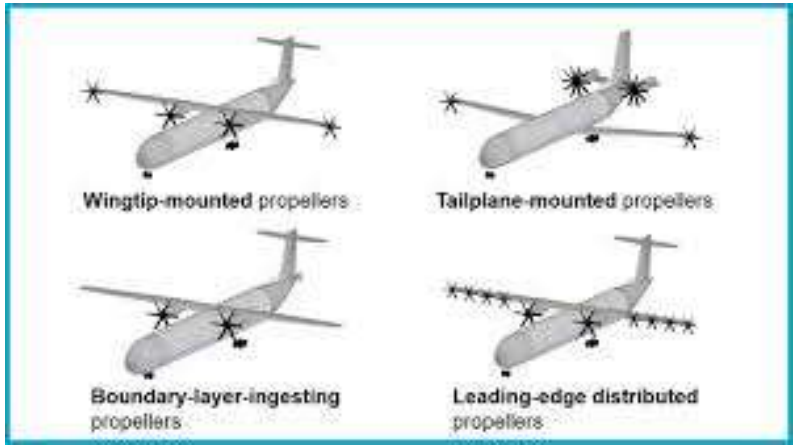




Distributed Propulsion

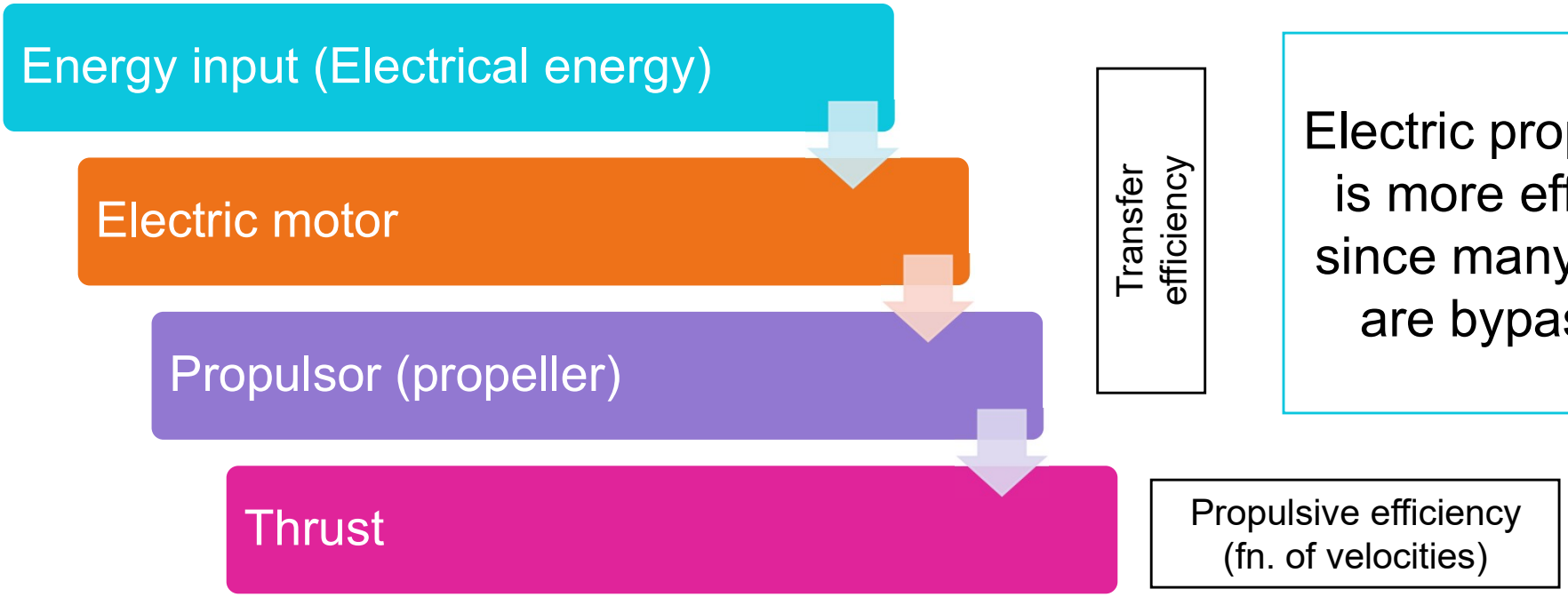
- Wingtip propellers $\Rightarrow D_L\downarrow$
  - Rotating opposite to wing tip vortices (30-40% of aircraft drag in cruise)
  - Introduce swirl leading to upwash on wing LE propellers

- LE distributed propellers $\Rightarrow D_L\downarrow$ & $D_0\downarrow$
  - Reduced wing size & weight
  - Reduced noise by synchrophasing the propellers



NASA Remakes the Airplane Wing With Electric Propulsion and a Whole Lot of Propellers - IEEE Spectrum

Electric propulsion efficiencies



Electric propulsion is more efficient since many steps are bypassed

Electric aircraft

- Battery + Electric Motor + Propeller
- Very high (~100%) efficiency
- Challenges ⇒ limited range
  - Battery wt. ~ 40 times of jet fuel for same energy
  - Battery stack ⇒ 10% addn. wt.
  - Residual charge ~ 10-20%
  - Low temp. performance ↓
  - High power performance ↓

Some myths & some facts

- Nuclear power is the silver bullet
  - Reactors too heavy due to shielding
  - Leakage in case of crash?
  - Green power on ground?
- Solar power is the panacea
  - Typically <1% of total power requirements
  - Night flights?

- **SOLAR IMPULSE -2**
  - **Energy:** 21 kW-h (76 MJ) Li-ion battery charged by solar cells
  - **Powerplant:** 4×7.5 kW electric motors
  - **Thrust producer:** Distributed propellers
- **US Navy upgrade- WIP**
  - Unmanned flight
  - 90 days endurance
  - Communication relay

University of BRISTOL

Lecture-7: Sustainable Propulsion & Fuel Systems

7C: Alternate Fuel Systems

Samudra Dasgupta

Senior Lecturer – CADE, University of Bristol

bristol.ac.uk

Conventional aviation fuel

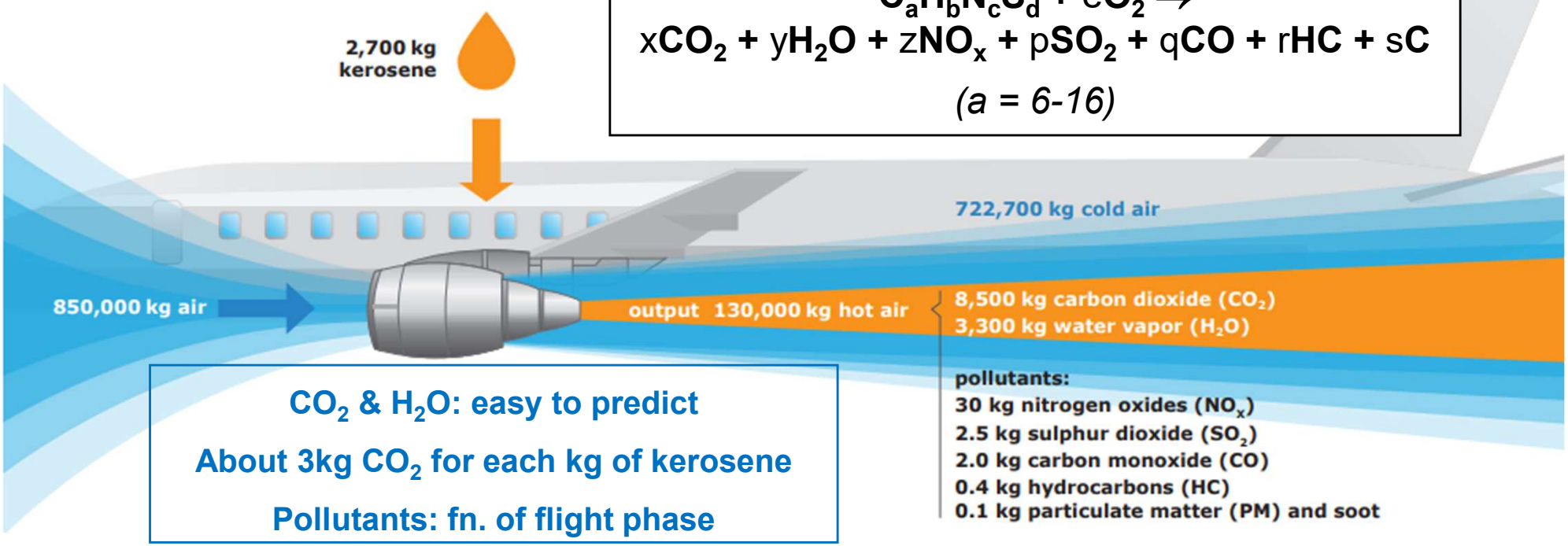
- **Jet A1**: suitable for flying at high altitudes (freezing point of −47°C)
- **Jet A**: similar to Jet A1, but with a higher freezing point of −40°C
- **Jet B**: fuel with a low freezing point of −76°C
- Essentially refined **Kerosene**
  - Petroleum distillate that's a mixture of hydrocarbons, including alkanes, alkyl benzenes, and naphthalenes (generally C12 to C15)

Kerosene combustion

Emissions from a typical two-engine jet aircraft during 1-hour flight with 150 passengers (Source: FOCA)

$$C_aH_bN_cS_d + eO_2 \rightarrow$$
$$xCO_2 + yH_2O + zNO_x + pSO_2 + qCO + rHC + sC$$

(a = 6-16)



CO_2 & H_2O: easy to predict

About 3kg CO_2 for each kg of kerosene

Pollutants: fn. of flight phase

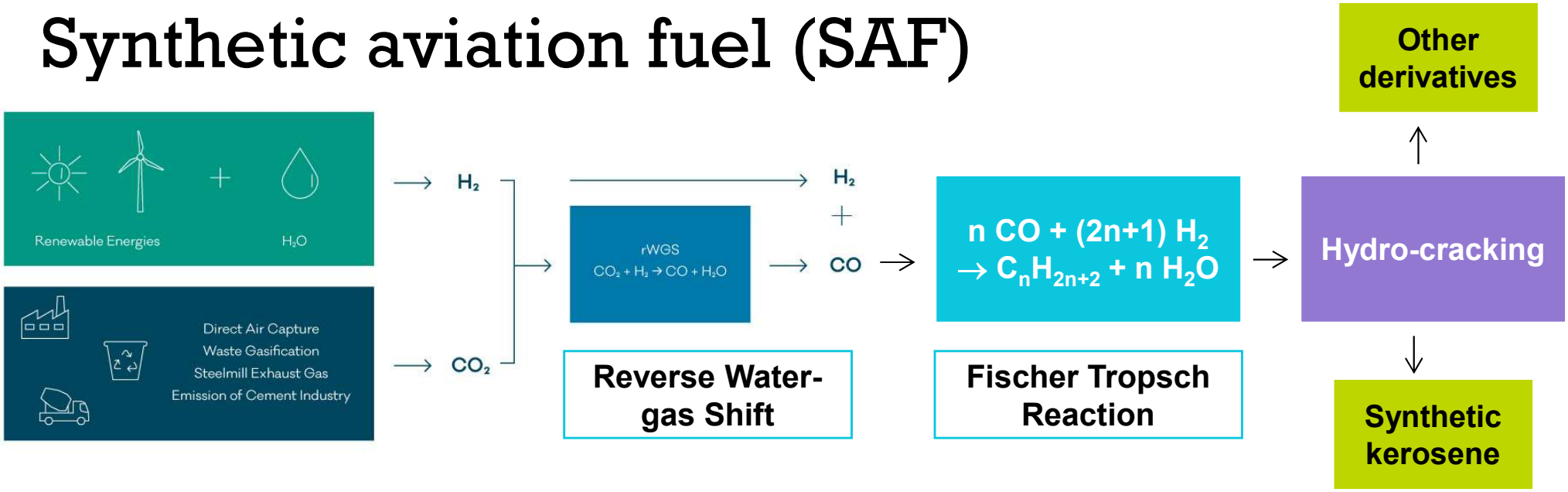
European Aviation Environmental Report 2016 -72dpi.pdf (europa.eu)

Case for & against kerosene

- For
  - Very high MED(43 MJ/kg)
  - Very high VED
  - Easily available (current consumption is about 1 million metric ton per day)
  - All current engines & airport infra is aligned
- Against
  - Unsustainable
    - Fossil fuel (unsustainable)
    - CO_2 & CO emissions
    - NO_x & SO_x emissions
  - Geopolitical concerns

Alternatives???

Synthetic aviation fuel (SAF)



- 10% SAF in aviation fuel mix by 2030 in UK, many a/c use it as on date
  - Normal liquid fuel (conventional infra & ops)
  - Composition is tunable based on catalyst, T, P, H2/CO ratio etc.
- Alternate: Bio-mass derived synthesized paraffin kerosene (Bio-kerosene)
  - Current capacity (20 million metric tonnes per year ~1/3rd of the requirement)

CORSIA life-cycle assessment for SAF

- CORSIA: The first internationally adopted approach to calculate life-cycle GHG emissions for aviation fuels

Core LCA [gCO_2e/MJ] =

$$e_{fc} + e_{fhc} + e_{fp} + e_{ft} + e_{fc} + e_{fut} + e_{fuc}$$

e_{fc} = feedstock cultivation
e_{fhc} = feedstock harvesting and collection
e_{fp} = feedstock processing
e_{ft} = feedstock transportation
e_{fc} = feedstock to-fuel conversion
e_{fut} = fuel transportation and distribution
e_{fuc} = fuel combustion in an aircraft engine

Any regional / seasonal variations up to 8.9 gCO_2e/MJ (10% of jet fuel baseline) is ignored, otherwise reasons investigated and harmonised

bristol.ac.uk

Contd…

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: www.elsevier.com/locate/rser

ELSEVIER

CORSIA: The first internationally adopted approach to calculate life-cycle GHG emissions for aviation fuels

Matteo Prussi [a], Uisung Lee [b,*], Michael Wang [b], Robert Malina [c], Hugo Valin [d], Farzad Taheripour [e], César Velarde [f], Mark D. Staples [g], Laura Lonza [h], James I. Hileman [i]

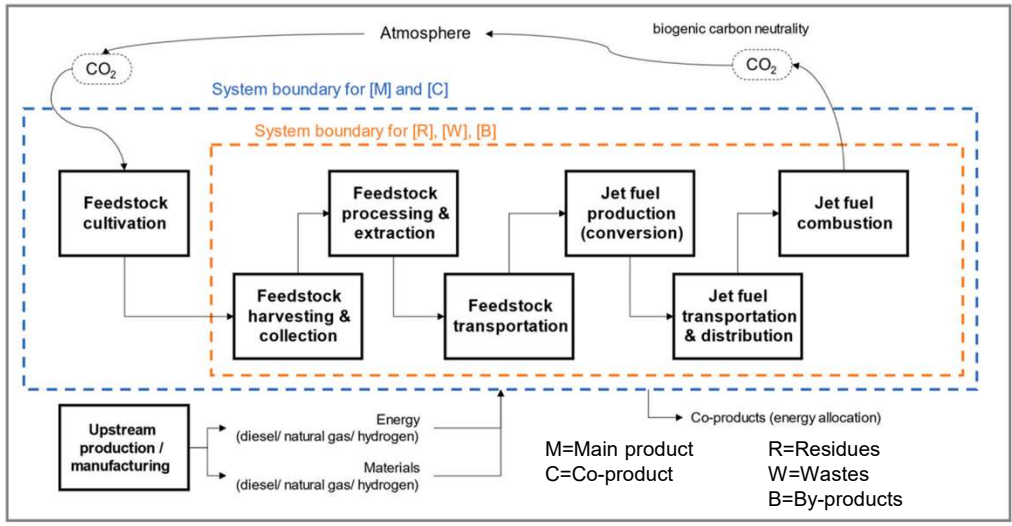


Fig. 1. The system boundary for core LCA of CORSIA SAFs.

CORSIA life-cycle assessment for SAF (contd.)

- Two databases/models have been used for evaluating core LCA values
  - E3 database (E3db)
  - Greenhouse gases, Regulated Emissions, and Energy use in Technologies (GREET®)

FT = Fischer-Tropsch
HEFA = hydro-processed esters & fatty acids
SIP = Synthesized iso-paraffins
Iso-buOH ATJ = Iso-butanol alcohol-to-jet
ETJ = Ethanol-to-jet

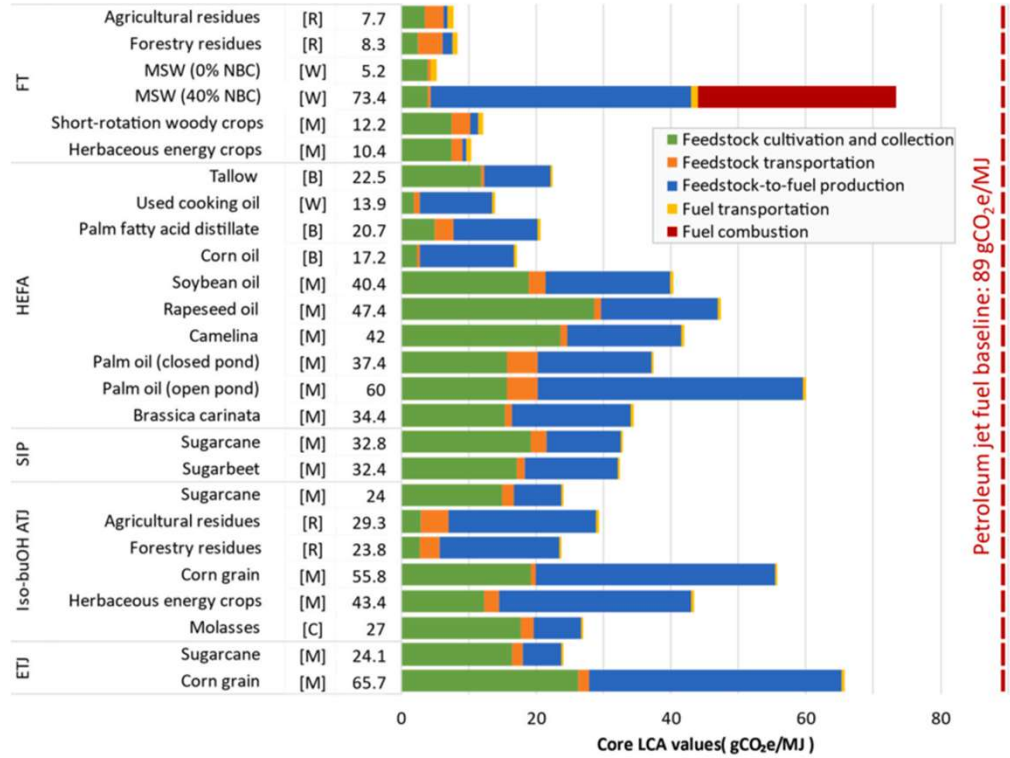


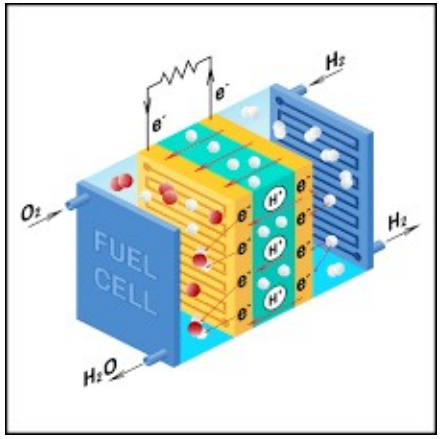
Fig. 2. Default core LCA values of SAF production pathways approved by ICAO to date. (NBC: non-biogenic carbon content).

Other possible alternatives

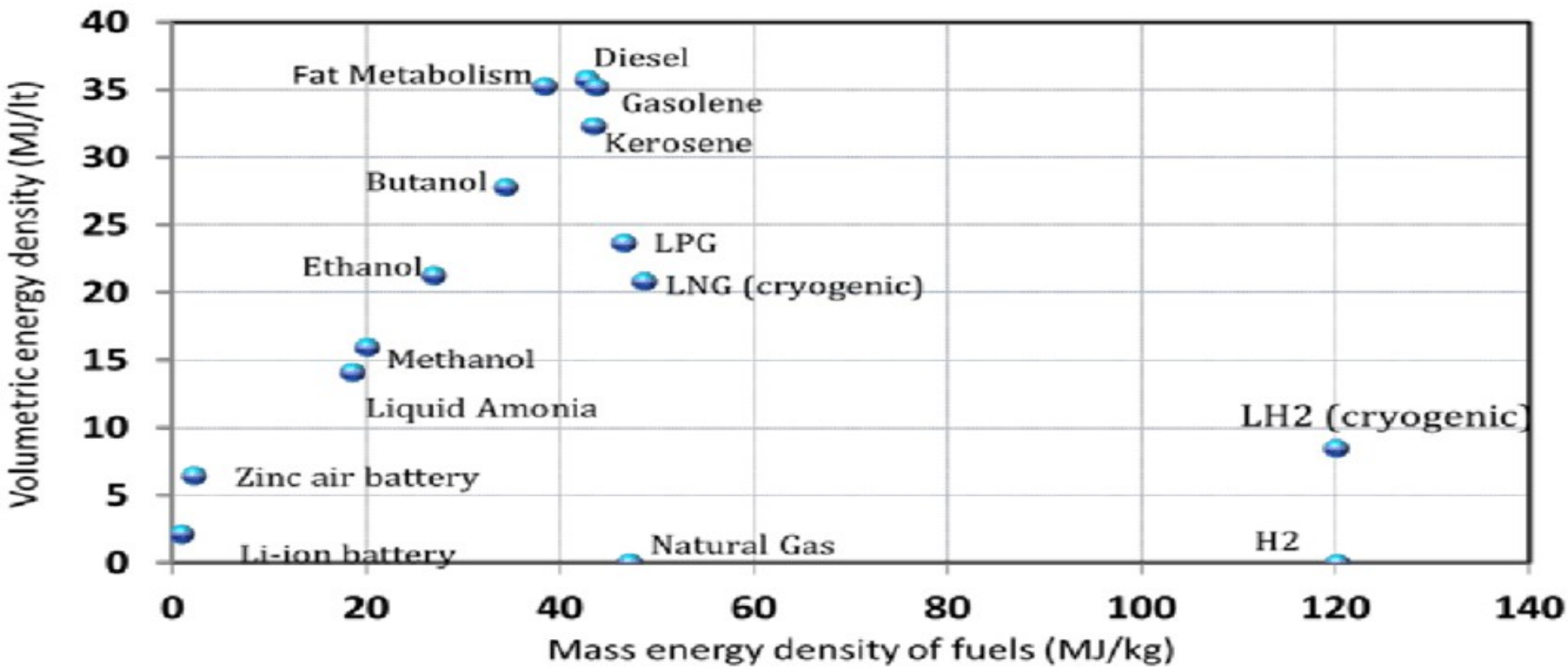
Green ammonia (l)	Alcohols (lower)	Synthetic green methane (l)
• Haber Bosch Process • 1bar/240K or 8bar/RT • Low ED • Corrosive & toxic	• Acceptable ED • Contains oxygen leading to soot & contrail • Toxic • No dumping / slippage	• From green H_2 & atm. CO2 • Low ED at 1bar/111K • Huge energy to liquify • 30X harmful GHG than CO_2 • No slippage

Hydrogen avatars

- H_2 (green) + atmospheric CO_2
  - Synthetic aviation fuel
- H_2 from fuel cell
  - Serious promise for smaller aircraft
- H_2 combustion engine
  - Proof of concept projects



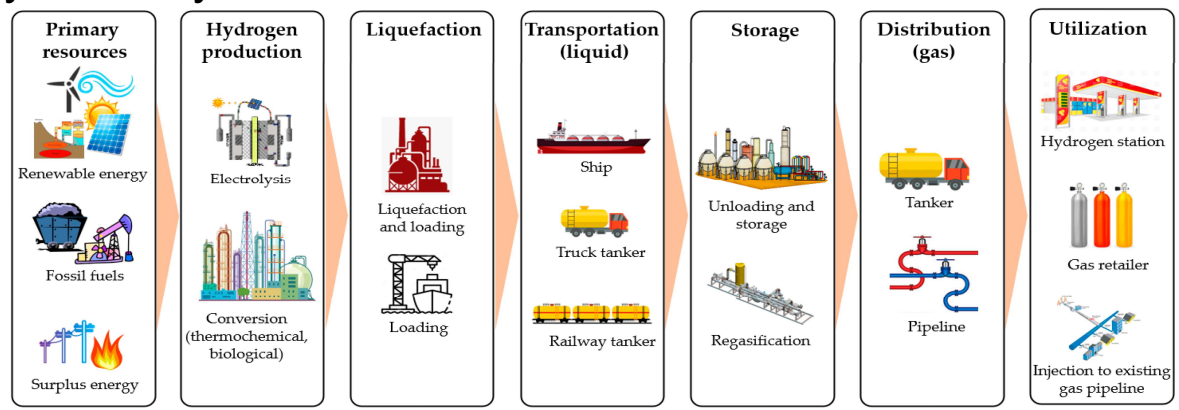
Comparison



(1) (PDF) Ahead: Paving the Way for Next Generation Aircraft and Engine

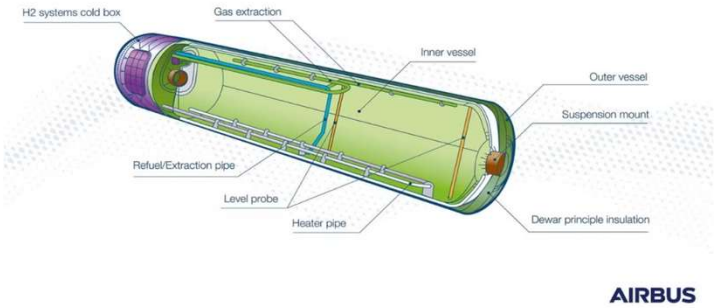
Challenges of H_2 economy

- Green H_2(g) from green electricity by electrolysis of water
  - Very low VED at NTP
- H_2 (l) @ 20K (better VED)
  - Cryogenic embrittlement
  - 20% energy consumed to liquify H_2
  - Cold chain constraints
  - Sloshing etc.
- H_2 (l) @ 750 bars & RT
  - Lower VED
  - Serious safety issues



Airbus ZEROe

- Modified A380 as FTB – stability, volume & propulsion redundancy
  - **Hydrogen combustion engine**
    - All regular engines with additional H2 engine on the fuselage
    - 2 fuselage frames reinforced and fitting external stub
    - Specially adapted pylon to attach the pod to the stub
    - Support high sustained static and dynamic loads
    - 4 special cryogenic tanks to contain the liq. H2 fuel
    - CFRP safety 'tent'
  - **Hydrogen fuel cell powered turbo-props**
    - Open fan engine under the wing (turbo prop look-alike)
    - Patented 'engine pod' containing hydrogen fuel cells
    - Designed for high-wing, 100-seat regional airliner



University of BRISTOL

Thank you

samudra.dasgupta@bristol.ac.uk

https://in.linkedin.com/in/samudradg

bristol.ac.uk